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OOP Lab 5

Static Data Members / Functions in Java

Task 1 - Student Class with Static University

```
class Student {
    String name;
    int rollNumber;
    static String university = "FAST University";

    Student(String name, int roll) {
        this.name = name;
        this.rollNumber = roll;
    }

    void displayInfo() {
        System.out.println("Name: " + name);
        System.out.println("Roll No: " + rollNumber);
        System.out.println("University: " + university);
        System.out.println("---");
    }

    static void changeUniversity(String newUni) {
        university = newUni;
    }
}

public class Task1_Student {
    public static void main(String[] args) {
        Student s1 = new Student("Ali", 101);
        Student s2 = new Student("Sara", 102);

        s1.displayInfo();
        s2.displayInfo();

        Student.changeUniversity("NUST University");
        System.out.println("After changing university:");
        s1.displayInfo();
        s2.displayInfo();
    }
}
```

Task 2 - Counter Class with Static Object Count

```
class Counter {
    String label;
    static int count = 0;
```

```
    Counter(String label) {
        this.label = label;
        count++;
    }

    static void displayCount() {
        System.out.println("Total objects created: " + count);
    }

    void display() {
        System.out.println("Object: " + label);
    }
}

public class Task2_Counter {
    public static void main(String[] args) {
        Counter.displayCount();

        Counter c1 = new Counter("First");
        Counter.displayCount();

        Counter c2 = new Counter("Second");
        Counter c3 = new Counter("Third");
        Counter.displayCount();

        c1.display();
        c2.display();
        c3.display();
    }
}
```

Task 3 - Employee Class with Static Company Name

```
class Employee {
    String name;
    int id;
    String department;
    static String companyName = "TechCorp Pvt Ltd";

    Employee(String name, int id, String dept) {
        this.name = name;
        this.id = id;
        this.department = dept;
    }

    void displayInfo() {
        System.out.println("Company: " + companyName);
        System.out.println("Name: " + name);
        System.out.println("ID: " + id);
        System.out.println("Dept: " + department);
        System.out.println("---");
    }

    static void changeCompany(String newName) {
        companyName = newName;
```

```
    }
}

public class Task3_Employee {
    public static void main(String[] args) {
        Employee e1 = new Employee("Ahmed", 1, "Engineering");
        Employee e2 = new Employee("Fatima", 2, "HR");

        e1.displayInfo();
        e2.displayInfo();

        Employee.changeCompany("NextGen Solutions");
        System.out.println("After company rebrand:");
        e1.displayInfo();
        e2.displayInfo();
    }
}
```

Task 4 - BankAccount Class with Static Interest Rate

```
class BankAccount {
    String ownerName;
    int accountNumber;
    double balance;
    static double interestRate = 5.0;

    BankAccount(String owner, int accNum, double balance) {
        this.ownerName = owner;
        this.accountNumber = accNum;
        this.balance = balance;
    }

    void displayDetails() {
        double interest = (balance * interestRate) / 100;
        System.out.println("Owner: " + ownerName);
        System.out.println("Account No: " + accountNumber);
        System.out.printf("Balance: Rs. %.2f%n", balance);
        System.out.printf("Interest Rate: %.1f%%%n", interestRate);
        System.out.printf("Annual Interest: Rs. %.2f%n", interest);
        System.out.println("---");
    }

    static void setInterestRate(double rate) {
        interestRate = rate;
    }
}

public class Task4_BankAccount {
    public static void main(String[] args) {
        BankAccount a1 = new BankAccount("Zain", 1001, 50000);
        BankAccount a2 = new BankAccount("Hina", 1002, 120000);

        a1.displayDetails();
        a2.displayDetails();

        BankAccount.setInterestRate(7.5);
```

```
        System.out.println("After rate increase:");
        a1.displayDetails();
        a2.displayDetails();
    }
}
```

Task 5 - Book Class with Static Library Name

```
class Book {
    String title;
    String author;
    int year;
    static String libraryName = "City Central Library";

    Book(String title, String author, int year) {
        this.title = title;
        this.author = author;
        this.year = year;
    }

    void displayInfo() {
        System.out.println("Library: " + libraryName);
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Year: " + year);
        System.out.println("---");
    }

    static void changeLibrary(String newLib) {
        libraryName = newLib;
    }
}

public class Task5_Book {
    public static void main(String[] args) {
        Book b1 = new Book("Java Basics", "James Gosling", 2010);
        Book b2 = new Book("Clean Code", "Robert Martin", 2008);
        Book b3 = new Book("Design Patterns", "Gang of Four", 1994);

        b1.displayInfo();
        b2.displayInfo();
        b3.displayInfo();

        Book.changeLibrary("National Digital Library");
        System.out.println("After library rename:");
        b1.displayInfo();
    }
}
```